INCH-POUND

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19 MAY 2009
SUPERSEDING
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DETAIL SPECIFICATION

CAN, WATER, MILITARY: PLASTIC, 5-GALLON CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 5-gallon capacity, rectangular, plastic, military water can with carrying handle.

1.2 Classification. The plastic 5-gallon water containers will be of the following types (see 6.2):

Type 1 – Tan
Type 2 – Green

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in section 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for Additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Philadelphia (DSCP), ATTN: DSCP-NASA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or e-mail to dscpg&inspecomments@dla.mil . Since contact information can change, you may want to verify the currency of this address information using ASSIST Online database at http://assist.daps.dla.mil .

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FEDERAL SPECIFICATION

L-P-390	- Plastic, Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium, and High Density)

FEDERAL STANDARDS

FED-STD-595/30279	- Brown, Flat, Lusterless
FED-STD-595/34094	- Green, Flat, Lusterless

(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Document Ordering Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other government documents, drawings, and publications. The following other government documents, drawings, publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-9-248	- Can, Water, Military, Plastic, 5-Gallon Capacity; Assembly
2-9-249	- Can, Water, Military, Plastic, 5-Gallon Capacity; Cap Assembly
2-9-285	- Adapter, Assembly and Details

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UXT, NATICK, MA 01760-5017.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	- Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from www.asq.org or the American Society for Quality, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained

3. REQUIREMENTS

3.1 First articles. When specified (see 6.2), a sample shall be subjected to the first article inspection (see 6.3) in accordance with 4.2.

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3.2 Federal Food and Drug Act certification. Prior to the approval of the first article, if one is submitted, or prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative, that the plastic for the water can and the antioxidants, light stabilizers, or other substances incorporated in the plastic proposed to be supplied under this specification meets the requirements of 21 CFR – Food and Drugs: Indirect Food Additives: Polymers, Subpart B, Part 177, Chapter I, Title 21, Code for Federal Requirements for Contact with Food. Acceptable evidence of meeting the requirements of this document shall be a certified test report from a recognized independent testing laboratory, acceptable to the Department of the Army Surgeon General. This certification shall be required for all acquisitions without exception.

3.3 Materials.

3.3.1 Polyethylene, high density. The plastic body shall consist of a high density polyethylene copolymer resin conforming to type I, class H of L-P-390 plus a light stabilizer, and requirements of 3.2 except that the basic resin prior to adding the light stabilizer shall have a density of not less than 0.954 and not more than 0.956, and a melt index of not more than 0.3.

3.3.2 Polyethylene, low density. The complete plastic cap and strap shall be low density virgin polyethylene conforming to type I, class L of L-P-390 plus a light stabilizer and requirements of 3.2.

3.4 Design and construction. The design and construction of the water container shall be as shown on the drawings specified in 2.2.2 and as specified herein. Silicone-type mold release agents shall not be used with polyethylene. Mold release agents which are not active stress-cracking agents for polyethylene shall be used. A certificate of compliance shall be furnished stating that the mold release agents comply with this requirement (see 4.4). A certificate of compliance shall be furnished stating that the material with the light stabilizers used on the complete cap and body meet the test requirements of 4.5.4 (see 6.4.2). The male and female threads shall be fully formed and free from flash or thread misalignment at the parting line. The finished water can shall not leak when tested as specified in 4.5.1.

3.4.1 Body. The body shall be fabricated from the high density polyethylene specified in 3.3.1, and shall consist of a bottom, four sides, and top with handle and a body neck and shall be formed by blow molding to give a one piece construction. The body walls shall be a minimum of 0.100 inch thick, except that the thickness of the four top outside curved surfaces of the handle may be 0.075 inch minimum; however, the 0.075 inch minimum thickness shall not extend over an area greater than 0.25 inch in diameter. The body neck shall have a thickness of not less than 0.151 inch in all areas below the area finished to form the liquid seal. The body walls shall not crack or break when tested as specified in 4.5.2. The upper surface of the body neck that comes in contact with the tapered sealing surface on the inside of the cap to form a seal shall be flat and smooth and may be out of round, in a plane parallel to the top surface of the body neck, provided that the fit or water tightness is not impaired. The interior surface of the body shall be smooth. The interior rounded surface area where the body and the neck join shall have a uniform texture having no rough areas, surface abnormalities, surface deformations, or surface irregularities.

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3.4.2 Cap. The cap assembly components, except the retaining pin, shall be injection molded and fabricated of the material specified in 3.3.2. The retaining pin shall be fabricated from the material specified in 3.3.1 and may be sealed to the breather stud cap using a heating method that does not leave burn marks. The water can cap shall lock in the closed or tightened position by means of a tab on the water can cap assembly which is engaged to the raised tab located at the base of the body pouring neck on the body. The breather stud cap shall be secured to the breather stud in the closed or tightened position by means of a retaining pin inserted through both the breather stud and breather stud cap. The breather stud shall be free and clear of all mold flashes. The caps or cap areas shall not crack or break when the water can is tested in accordance with 4.5.2. The plastic caps shall not crack when tested in accordance with 4.5.3. The upper surface of the cap pouring spout neck, that comes in contact with the tapered sealing surface on the inside of the pouring spout cap to form a seal, shall be flat and smooth and may have an out of roundness in a plane parallel to the top surface of the pouring spout provided that the fit or water tightness is not impaired.

3.5 Color. The through color of the material for the body, caps, and breather after molding shall approximate color No. 30279 (sand, lusterless) for Type I water cans and color No. 34094 (green, lusterless) for Type II water cans of FED-STD-595. The complete cap and breather shall match the shade of the body.

3.6 Finish. The exterior surface of the water can and the cap assembly shall have a smooth and lusterless finish throughout. All surfaces shall be free of dirt, dust, and foreign matter inclusion. The finish shall be produced from a die that has cavity surfaces finished in vapor blast, water hone, or similar satin finish, but not by the application of any protective coating, lacquer, or other materials. The inside of the cap assembly, including threaded portion, shall possess a uniformly high luster throughout as imparted by a polished die.

3.7 Marking. The water can shall be permanently marked “Property of the U.S. Government” and “WATER” by an embossed molded-in process. The water can shall be marked by an embossed molded-in process with the year of manufacture and the contractor’s name or trademark. The container cap shall be marked with raised characters by an embossed molded-in process with the manufacturer’s name or trademark.

3.8 Workmanship. The water can shall be clean, void of plastic shavings, and free from flash, bubbles, cuts, tears, holes, burns, breaks, cracks, pinholes, pins, warpage, blisters, scratches, or any weld marks or sink marks, except that mold parting lines are acceptable. The water can shall not show any signs of discoloration, surface deterioration, roughness, orange peel, or grease.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

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4.2 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.3.2.1 and the specified dimensions, and tested in accordance with 4.5.1, 4.5.2, 4.5.3, and 4.5.4. Failure to pass any test shall be cause for rejection of the first article. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.3 Conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with ASQ Z1.4.

4.3.1 Component and material inspection. Components and materials shall be inspected by the contractor in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 In-process examination.

4.3.2.1 In-process examination. Examination shall be made of the fabrication processes in table I to establish conformance with specified requirements. Whenever noncompliance is noted, correction shall be made to the applicable process and all items processed.

Table I. In-process examination.

Requirement	Reference paragraph
Blow molding of can body	3.4 and 3.4.1
Injection molding	3.4.2
Cavity surfaces of finished die	3.6

4.3.3 End item inspection. The inspection lot shall be all water cans offered for inspection at one time. The sample unit shall be one complete water can.

4.3.3.1 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of cans. The sample unit shall be one can. The inspection level shall be II and the acceptable quality limit (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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Table II. End item visual defects.

Examine	Defect	Classification Major	Classification Minor
Finish	Color not as specified.	101	
	Exterior surface of body, cap, or strap not smooth and not lusterless throughout (except any die parting lines).	102	
	Inside surface of cap, including threaded portion, not smooth.		201
	Inside surface of cap, including threaded portion, not of uniform high luster.		202
Construction and workmanship (applicable to all assemblies and Components)	Component missing	103	
	Any bubbles, cuts, tears, holes, flash, burns, breaks, cracks, pinholes, pits, warpage, blisters, scratches, welds or sink marks, or not clean or smooth.	104	
	Any discoloration, surface deterioration or roughness, foreign inclusion, or orange peel.	105	
Body	Sealing surface of lip is not flat, smooth, and free from sink marks or depressions.	106	
	Interior of neck opening not smooth.	107	
	Interior surfaces of the body not smooth.	108	
Cap and strap (large cap or spout as applicable)	Sealing surface in cap is irregular, not smooth, or contains high or low points.	109	
	Strap locking tab does not engage with tab on body in closed position.	110	
	Cap does not screw on body neck or pouring spout.	111	
	Cap binds when screwed on body neck or pouring spout.		203
	Strap does not engage in retention groove		204
	Breather air vent holes not clear.	112	
Threads	Threads in caps, body neck, pouring spout, or breather stripped.	113	
	Threads not fully formed, not free from flash, or misaligned at parting line.	114	
Marking	Missing, incomplete, not legible, not specified type or size, or not in proper location.		205

4.3.3.2 End item dimensional examination. The end items shall be examined for conformance to the dimensions annotated with an asterisk on drawing 2-9-248. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of cans. The sample unit shall be one can. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.3.3.3 End item testing. The end items shall be tested as specified in 4.5.1, 4.5.2, and 4.5.3 to detect the defects specified in table II. The lot size shall be expressed in units of cans. The sample unit shall be can. The inspection level shall be S-3, and the AQL, expressed in terms of defects per hundred units, shall be 2.5. Any test failure shall be cause for rejection of the lot.

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4.3.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container
Content	Number per container is more or less than required.

4.3.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finish dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Load not bonded with required straps specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.4 Certification examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the document requirements shall be examined and validated for mold release agents and light stabilizers (see 3.2 and 3.4).

4.5 Test methods.

4.5.1 Leakage test. The water can shall be filled with water at 73° F ± 3° F. The cap shall be screwed down and tightened, and the water can shall be allowed to remain on its bottom for 10 minutes and examined for compliance with 3.4. The exterior of the water can and cap shall be free of any moisture. The water can shall be rotated 180° so as to be resting on its handle, with the cap facing down for not less than one-half hour and examined for any leakage or presence of moisture. To determine compliance with 3.4, any evidence of leaking or presence of moisture shall constitute failure.

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4.5.2 Drop test. The water can shall be filled with 5 gallons of water at 73° F ± 3° F. The filled water can shall be conditioned for 24 hours at 0° F. The conditioned water can shall then be dropped from a height of 6 feet onto a solid concrete horizontal surface in such a manner that the base of the container strikes squarely against the surface to determine compliance with soundness requirements in 3.4. After the above drop test has been accomplished, the conditioned water can shall be allowed to regain room temperature and then examined for leakage as specified in 4.5.1. Any evidence of leakage shall constitute failure of this test.

4.5.3 Environmental stress cracking tests for caps. The neck threads on the body and the cap threads, including the area between the base of threads and sealing ring, shall be thoroughly coated with full strength Igepal CO-630 (see 6.5.1). The cap shall be screwed down on the body and tightened to a torque of not less than 15 foot pounds by the use of a torque wrench utilizing the adapter shown on drawing 2-9-285. The container shall then be placed in an upright position in a circulating air area at a temperature of 140° F ± 2° F. After 30 hours exposure, the container shall be removed from the oven and the cap examined for cracks. If the cap shows no cracks, recheck the torque. If it is less than 15 foot pounds, retighten the cap to a minimum of 15 foot pounds and re-examine the cap immediately for cracks. The pouring cap shall be screwed to the pouring spout and the breather shall be open during the period of the test. Any visible evidence of cracks or cracking shall constitute failure of this test.

4.5.4 Light stabilizer tests for materials used in the body and complete cap. Plastic samples (compression molded plaques) shall be exposed to a 6,000 watt Xenon Arc Weatherometer for 130, 24-hour days, straight running time. There shall be four one-half hour moisture cycles a day. At the completion of the tests, the samples shall show no signs of deterioration or degradation.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or inhouse contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Services or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The water cans specified herein are used for supplying potable water to personnel of the Armed Services in zones I through VI, but not in zone VII (Arctic).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of water can desired (see 1.2).
c. When a first article is required (see 3.1, 4.2, and 6.3).
d. Packaging requirements (see 5.1).

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6.3 First article. When a first article is required, it will be inspected and approved under the appropriate provision of Federal Acquisition Regulation (FAR) 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Commercial sources.

6.4.1 GAF Corporation. Igepal CO-630, required in 4.5.3, may be obtained from GAF Corporation (CAGE 24299), 1361 Alps Road, Wayne, NJ 07470.

6.4.2 Ciba Geigy Corporation. Tinuvin 622, a product of Ciba Geigy Corporation, Plastics and Additive Division, 7 Skyline Drive, Hawthorne, NY 10532, has been found to meet the requirements of 3.4.

6.5 Subject term (keyword) listing.

Container
Container, polyethylene
Jug
Polyethylene container

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army – GL
Navy – SA
Air Force – 99

Preparing activity:
DLA – IS

(7240-2009-001)

Review activities:
Army – MD
Navy – CG, MC, YD
Air Force – 84

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